Timeline: Adding the time dimension to spreadsheets

ANONYMOUS AUTHOR(S)

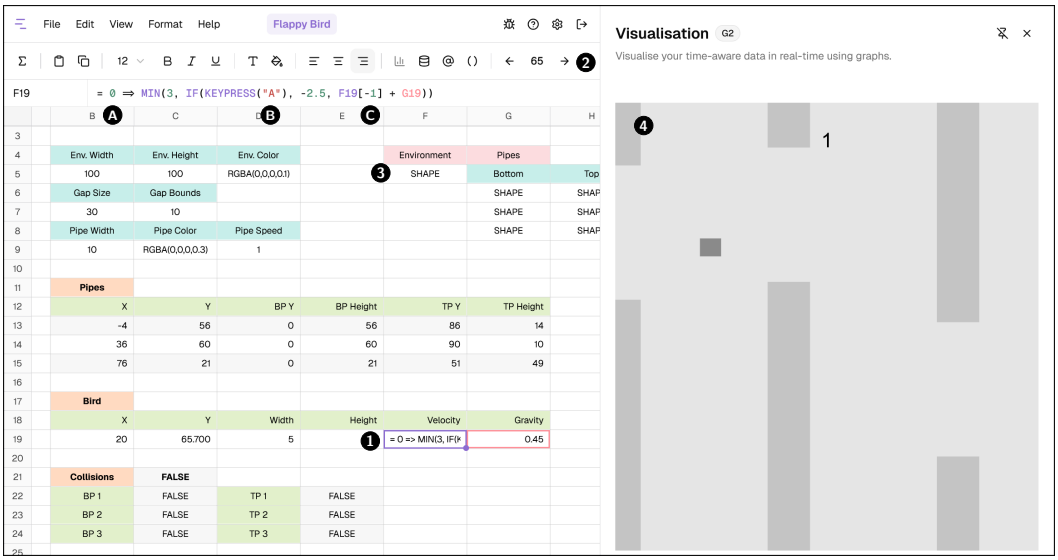

Fig. 1. A flappy bird game clone in Timeline. ❶ The selected formula computes the Y velocity in the current step. ❷ The system uses discrete time. ❸ Visualizations are built using a composable grammar of graphics and ❹ can be displayed inline or in a separate panel. The formula for the velocity Ⓐ specifies an initial value, Ⓑ reacts to user-interface events, and Ⓒ accesses a previous value to compute the current one.

Spreadsheets make it easy to express computations over two-dimensional data, but two dimensions are not enough to express rich computations with time such as simulations, complex systems models, analyses of historical data, or interactive systems. We present Timeline, a system that adds discrete time to spreadsheets.

The remarkable insight from our work is that a large number of advanced programming language concepts can be directly applied in the context of spreadsheets with time. Timeline draws from *dataflow languages* to express computations over time, *coeffect systems* to store only the necessary previous values, *functional reactive programming* to support interactivity, *grammars of graphics* to support composable visualizations, and *typed holes* for inserting cell references in the formula editor.

In this paper, we provide an overview of the Timeline design and discuss how it adapts the aforementioned programming language innovations for the context of spreadsheets. We then formalize the evaluation of spreadsheets with discrete time through a *core calculus*, describe a coeffect-based inference for computing the required number of past values and prove that the optimization is sound. More broadly, this paper shows that established programming language ideas can often be productively used outside of their original domain.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 1557-735X/2026/8-ART111
https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Spreadsheets are sometimes referred to as "the most widely used programming systems in the world" [2]. Their uses range from personal productivity tools and decision making support systems to scientific analytical instruments and financial reporting systems [10]. However, spreadsheets are first-order, limited to tabular data structures, and lack abstraction mechanisms.

Many extensions to the core spreadsheet model have been proposed, generalising the tabular structure to more general forms [3, 7], adding support for objects [17], or incorporating support for streaming data to use spreadsheets as the back-end for web development [8, 9]. Abstraction mechanism developed for spreadsheets include sheet-defined functions [15, 20] and gridlets [14], while programming by example [11] simplifies data wrangling within spreadsheets.

As the above extensions show, extending spreadsheets is a difficult balancing act. How can we make spreadsheets more expressive, while preserving their liveness, directness and usability [13]? The aim of the Timeline project is to extend spreadsheets with a built-in notion of discrete time, making it possible to use spreadsheets in a range of domains that are difficult to tackle with only two dimensions [16] such as physics simulations, agent-based modelling [1], financial data analysis, or simple interactive applications. As illustrated in Figure 1, Timeline adds an implicit discrete time to the spreadsheet programming model and extends the formula language to allow references to previous values and user interface events, but otherwise preserves the standard spreadsheet model.

Although Timeline is interesting as an exploration of the expressive power of spreadsheets with discrete time, this paper considers it primarily from the technical perspective of programming language design and implementation. We show that a large number of concepts and techniques developed in the context of programming language research are directly applicable to spreadsheets.

The key addition to Timeline is a built-in notion of discrete time. Formulas in cells can refer to other cells (e.g. `G19`), but also values computed in earlier steps (e.g. `F19[-1]`). To support this, the runtime and language of Timeline draws inspiration from research on *synchronous dataflow languages* [4, 12]. To ensure that simulations and interactive applications (e.g., Figure 1) execute using bounded memory when possible, we use a program analysis based on *coeffects* [18, 19]

to

to

To

Contributions. This paper presents the design and implementation of a spreadsheet system with discrete time and captures its essence through a core calculus.

- We provide an overview of the Timeline system design through a simple physics simulation example (§2) and then examine the expressive power of the programming model through a number of case studies covering agent-based modelling, simulations and finance (§5).
- We discuss how how the design and implementation of Timeline adapts a range of advanced programming language concepts and techniques for the context of spreadsheet systems (§3).
- We present a core calculus of the Timeline execution model, which captures the way previous values are stored during the spreadsheet evaluation (§4). We formalize a space optimization based on coeffects implemented in Timeline and prove its soundness (§4.1).

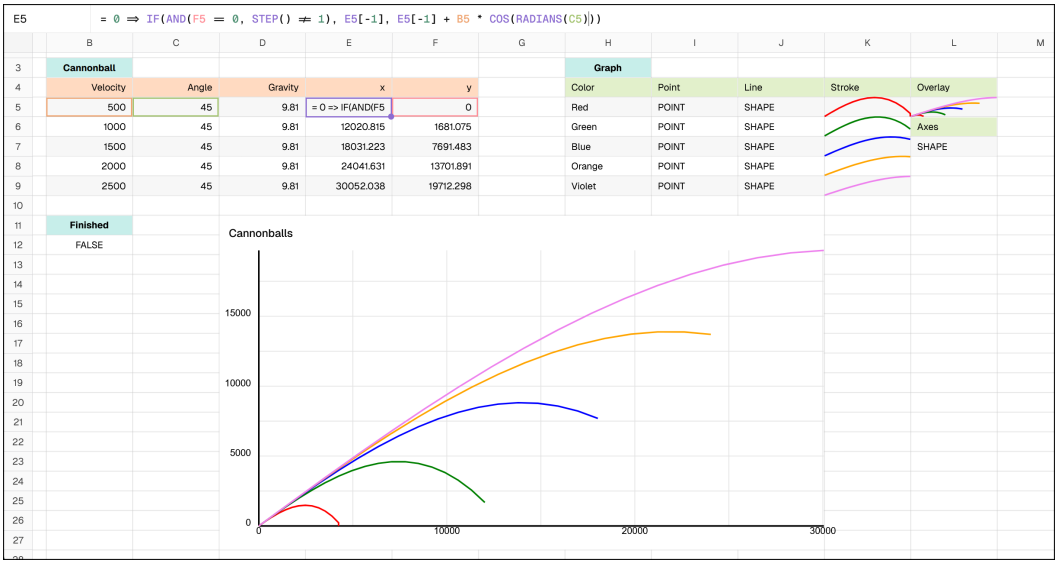

Fig. 2. A flappy bird game clone in Timeline. ❶ The selected formula computes the Y velocity in the current step. ❷ The system uses discrete time. ❸ Visualizations are built using a composable grammar of graphics and ❹ can be displayed inline or in a separate panel. The formula for the velocity Ⓐ specifies an initial value, Ⓑ reacts to user-interface events, and Ⓒ accesses a previous value to compute the current one.

2 Walkthrough

3 Language features

3.1 Synchronous languages

3.2 Coeffects

3.3 Functional reactive programming

3.4 Composable data vis

3.5 Typed holes

4 Calculus

4.1 Coeffect soundness

5 Case studies

5.1 Formative interviews

6 Related work

sheet defined functions [15] [20] and their semantics [6]

gridlets [14] and their semantics [21] [5]

References

[1] Sameera Abar, Georgios K. Theodoropoulos, Pierre Lemarinier, and Gregory M.P. O'Hare. 2017. Agent Based Modelling and Simulation tools: A review of the state-of-art software. *Computer Science Review* 24 (2017), 13–33. doi:10.1016/j.cosrev.2017.03.001

[2] Robin Abraham, Margaret Burnett, and Martin Erwig. 2009. *Spreadsheet Programming*. John Wiley & Sons, Ltd, 2804–2810. doi:10.1002/9780470050118.ecse415

[3] Pavel Bažant and Michaela Maršálková. 2018. A Non-Tabular Spreadsheet with Broad Applicability. In *Companion Proceedings of the 2nd International Conference on the Art, Science, and Engineering of Programming (Programming '18).*

Association for Computing Machinery, New York, NY, USA, 161–165. doi:10.1145/3191697.3214343

[4] A. Benveniste, P. Caspi, S.A. Edwards, N. Halbwachs, P. Le Guernic, and R. de Simone. 2003. The synchronous languages 12 years later. *Proc. IEEE* 91, 1 (2003), 64–83. doi:10.1109/JPROC.2002.805826

[5] Alexander Asp Bock, Thomas Bøgholm, Peter Sestoft, Bent Thomsen, and Lone Leth Thomsen. 2022. On the cost semantics for spreadsheets with sheet-defined functions. *J. Comput. Lang.* 69 (2022), 101103. doi:10.1016/J.COLA.2022.101103

[6] Alexander Asp Bock, Thomas Bøgholm, Peter Sestoft, Bent Thomsen, and Lone Leth Thomsen. 2020. On the semantics for spreadsheets with sheet-defined functions. *Journal of Computer Languages* 57 (2020), 100960. doi:10.1016/j.cola.2020.100960

[7] MARGARET BURNETT, JOHN ATWOOD, REBECCA WALPOLE DJANG, JAMES REICHWEIN, HERKIMER GOTTFRIED, and SHERRY YANG. 2001. Forms/3: A first-order visual language to explore the boundaries of the spreadsheet paradigm. *Journal of Functional Programming* 11, 2 (2001), 155–206. doi:10.1017/S0956796800003828

[8] Kerry Shih-Ping Chang and Brad A. Myers. 2014. Creating Interactive Web Data Applications with Spreadsheets. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology (UIST '14)*. Association for Computing Machinery, New York, NY, USA, 87–96. doi:10.1145/2642918.2647371

[9] Kerry Shih-Ping Chang and Brad A. Myers. 2015. A Spreadsheet Model for Handling Streaming Data. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. Association for Computing Machinery, New York, NY, USA, 3399–3402. doi:10.1145/2702123.2702587

[10] Thomas A Grossman and Vijay Mehrotra. 2023. A Use Case-Engineering Resources Taxonomy for Analytical Spreadsheet Models. In *21st EuSpRIG Annual Conference "The Spreadsheet Crisis: Regaining Control"*.

[11] Sumit Gulwani, William R. Harris, and Rishabh Singh. 2012. Spreadsheet Data Manipulation Using Examples. *Commun. ACM* 55, 8 (Aug. 2012), 97–105. doi:10.1145/2240236.2240260

[12] N. Halbwachs, P. Caspi, P. Raymond, and D. Pilaud. 1991. The synchronous data flow programming language LUSTRE. *Proc. IEEE* 79, 9 (1991), 1305–1320. doi:10.1109/5.97300

[13] Felienne Hermans, Bas Jansen, Sohon Roy, Efthimia Aivaloglou, Alaaeddin Swidan, and David Hoepelman. 2016. Spreadsheets are Code: An Overview of Software Engineering Approaches Applied to Spreadsheets. In *2016 IEEE 23rd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, Vol. 5. 56–65. doi:10.1109/SANER.2016.86

[14] Nima Joharizadeh, Advait Sarkar, Andrew D. Gordon, and Jack Williams. 2020. Gridlets: Reusing Spreadsheet Grids. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–7. doi:10.1145/3334480.3382806

[15] Simon Peyton Jones, Alan Blackwell, and Margaret Burnett. 2003. A user-centred approach to functions in Excel. In *Proceedings of the Eighth ACM SIGPLAN International Conference on Functional Programming* (Uppsala, Sweden) *(ICFP '03)*. Association for Computing Machinery, New York, NY, USA, 165–176. doi:10.1145/944705.944721

[16] Steve Litt. 2017. Mitigating Spreadsheet Risk in Complex Multi-Dimensional Models in Excel. In *Proceedings of the EuSpRIG 2017 Conference "Spreadsheet Risk Management"*. European Spreadsheet Risks Interest Group (EuSpRIG), London, UK.

[17] Matt McCutchen, Shachar Itzhaky, and Daniel Jackson. 2016. Object Spreadsheets: A New Computational Model for End-User Development of Data-Centric Web Applications. In *Proceedings of the 2016 ACM International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! 2016)*. Association for Computing Machinery, New York, NY, USA, 112–127. doi:10.1145/2986012.2986018

[18] Tomas Petricek, Dominic Orchard, and Alan Mycroft. 2013. Coeffects: Unified Static Analysis of Context-Dependence. In *Automata, Languages, and Programming*, Fedor V. Fomin, Rūsiņš Freivalds, Marta Kwiatkowska, and David Peleg (Eds.). Springer Berlin Heidelberg, 385–397.

[19] Tomas Petricek, Dominic Orchard, and Alan Mycroft. 2014. Coeffects: a calculus of context-dependent computation. In *Proceedings of the 19th ACM SIGPLAN International Conference on Functional Programming* (Gothenburg, Sweden) *(ICFP '14)*. Association for Computing Machinery, New York, NY, USA, 123–135. doi:10.1145/2628136.2628160

[20] Peter Sestoft and Jens Zeilund Sørensen. 2013. Sheet-Defined Functions: Implementation and Initial Evaluation. In *End-User Development*, Yvonne Dittrich, Margaret Burnett, Anders Mørch, and David Redmiles (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 88–103.

[21] Jack Williams, Nima Joharizadeh, Andrew D. Gordon, and Advait Sarkar. 2020. Higher-Order Spreadsheets with Spilled Arrays. In *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020 (Lecture Notes in Computer Science, Vol. 12075)*, Peter Müller (Ed.). Springer, 743–769. doi:10.1007/978-3-030-44914-8_27